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EXPLORATORY ANALYSIS OF EEG SPECTRAL FEATURES FOR PSYCHIATRIC SCREENING

Abstract: this study examines an open EEG dataset to assess the opportunities and limitations of spectral features for preliminary psychiatric screening. Demographic and diagnostic distributions were analysed before restricting the sample to participants aged 18–30 years. Band-power differences were then described relative to healthy controls, and two anomaly detectors were evaluated as an initial feasibility check. The analysis revealed pronounced sample imbalance and heterogeneous spectral changes across diagnoses. These findings define the limits of the empirical reference group and justify further testing of supervised models, without establishing diagnostic biomarkers.

Keywords: electroencephalography, exploratory data analysis, spectral power, empirical reference, anomaly detection.

Introduction

EEG records the electrical activity of the brain through electrodes placed on the scalp. Spectral analysis represents this activity in frequency bands associated with different functional states. In psychiatry, these measurements are investigated as additional sources of information rather than substitutes for clinical assessment [1, 2]. However, a visible difference between diagnostic groups does not necessarily provide a reliable rule for classifying an individual.

This study addresses two related objectives: to characterize the dataset and to describe spectral differences within a selected age group, and to assess whether deviations from a healthy reference can support binary screening. Exploratory analysis is central to both objectives: it

determines which participants can be compared and which conclusions the data can support.

Dataset and demographic analysis

The open EEG Psychiatric Disorders Dataset contains 945 participants and pre-calculated spectral and coherence variables. The source study describes five-minute, eyes-closed resting-state recordings [1]. The present analysis uses absolute power (AB): six bands at 19 electrodes, or 114 features. Coherence measures relationships between electrodes and is reserved for later work because this study focuses on band power.

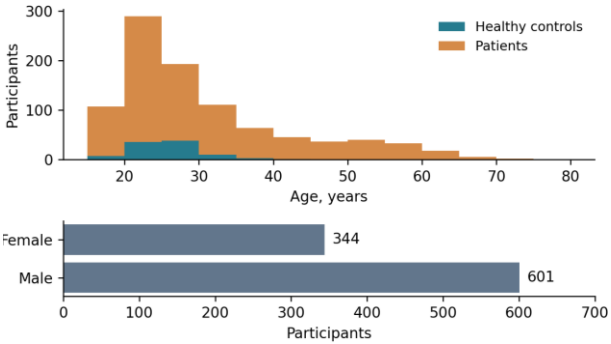


Figure 1 – Age distribution by clinical status and sex distribution in the full dataset

The sample includes 601 men (63.6%) and 344 women (36.4%). Mean age is 30.6±11.8 years. Healthy controls have a narrower age distribution than patients and a mean age of 25.7±4.5 years. Figure 1 therefore identifies a limitation of a single reference applied across the entire sample, not merely a demographic characteristic.

There is no universal numerical EEG norm that is independent of age and recording conditions [2]. Restricting the analysis to participants aged 18–30 years narrows the age range and retains 84 of the 95 healthy controls. However, this restriction does not eliminate age or sex confounding; these factors remain methodological limitations rather than being resolved by excluding them from model inputs.

Age-restricted sample and analysis strategy

The selected sample contains 599 participants: 84 healthy controls (14.0%) and 515 patients (86.0%). Figure 2 further demonstrates substantial imbalance across diagnostic categories. Depression is the largest patient group (123 participants), followed by behavioural addiction (78) and schizophrenia (65). Adjustment disorder and post-traumatic stress disorder are represented by only 21 and 12 observations, respectively.

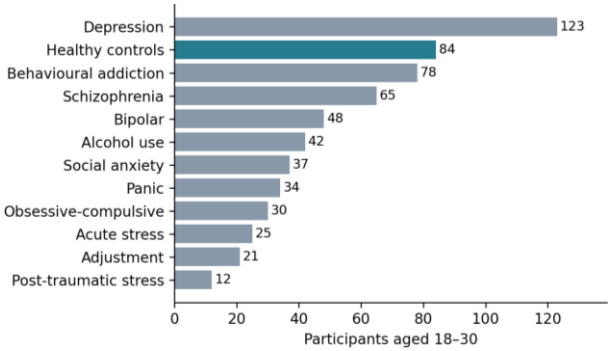


Figure 2 – Diagnostic composition of the sample aged 18–30 years

This distribution has two consequences. First, an algorithm that labels everyone as a patient already reaches approximately 86% accuracy; this is not evidence of useful screening. Second, estimates for small diagnostic groups are less stable. All diagnoses were retained for the binary model check, but the descriptive spectral comparison was limited to groups with at least 25 participants. This pragmatic cutoff does not guarantee statistical power.

The healthy spectral reference was defined using the clinical labels, without additionally selecting controls according to the autoencoder output. Otherwise, the reference would depend on the model being evaluated and could artificially narrow the apparent range of normal variation.

Spectral differences between diagnostic groups

For each participant, absolute power was averaged across the 19 electrodes separately within each frequency band. Group means were then compared with the corresponding healthy-control mean. The percentage difference was calculated as $D=100\times(P_{group}-P_{healthy})/P_{healthy}$. This quantity represents a relative change in absolute power, not relative band power expressed as a fraction of total EEG power.

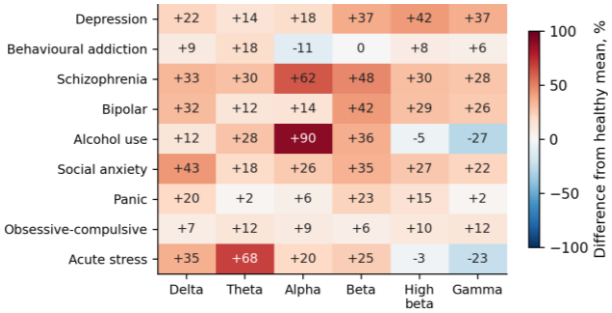


Figure 3 – Percentage differences in mean band power from healthy controls; descriptive estimates, not significance tests

Compared with healthy controls, the alcohol use disorder group exhibited higher mean alpha power (+90.0%) and lower gamma power (−27.4%) than controls. The depression group exhibited higher beta (+36.9%) and high-beta power (+42.2%); schizophrenia showed higher alpha (+62.3%) and beta power (+47.7%). Acute stress showed higher theta power (+68.5%). Because these patterns overlap across diagnoses, they should be interpreted as descriptive sample estimates rather than diagnostic signatures.

These group summaries do not show how accurately individual participants can be separated. Arithmetic means also remain sensitive to extreme values, and averaging across electrodes removes spatial detail. The heatmap is therefore an exploratory description that motivates the model evaluation rather than as evidence of diagnosis-specific spectral patterns.

Initial anomaly-detection check

Two anomaly detection models were compared to distinguish healthy participants from patients using EEG spectral features. Both learned a reference from healthy participants aged 18–30. Isolation Forest identifies unusual profiles through random partitions [3], whereas the autoencoder detects deviations through reconstruction error [4]. Neither model was trained to identify specific diagnoses.

Evaluation used ten stratified splits, with approximately 60% of observations allocated to training, 20% to calibration and 20% to testing. Standardisation and model fitting used only healthy training observations. Isolation Forest contained 500 trees; the autoencoder used hidden layers of 64–32–16–32–64 neurons, ReLU activation and Adam optimisation, with a maximum of 2,000 iterations.

To prioritise patient detection, thresholds were selected on the calibration subset by maximising specificity while maintaining sensitivity of at least 0.90. This was an experimental target, not a clinical standard. The test subset was used only for evaluation. Table 1 summarises the mean results.

Table 1 – Mean test performance across ten splits

Metric	Isolation Forest	Autoencoder
Sensitivity	0.911	0.898
Specificity	0.147	0.194
Balanced accuracy	0.529	0.546
ROC AUC	0.599	0.638

Isolation Forest achieved slightly higher mean sensitivity, whereas the autoencoder performed better in specificity, balanced accuracy and ROC AUC. However, both models incorrectly flagged most healthy controls. The autoencoder's higher mean ROC AUC therefore suggests only a modest advantage; statistical significance was not established. Overall, these findings show the limitations of detecting psychiatric conditions solely as deviations from a healthy reference and motivate further comparison with supervised methods.

Discussion and conclusion

The analysis characterised sample imbalance, established an age-restricted reference and quantified descriptive spectral differences. The initial model check showed that deviations from this reference do not reliably separate healthy controls from patients. Group-level spectral changes should therefore not be equated with individual diagnostic markers.

Interpretation is constrained by the small healthy-control group, the use of single-source dataset, unequal diagnostic-group sizes, and the absence of adjustment for medication, comorbidities and sex. The public table also prevents reprocessing raw EEG. These restrictions cannot be overcome simply by increasing model complexity.

These findings provide a rationale for evaluating whether supervised models can exploit multivariate feature combinations more effectively. Future comparisons require consistent data splits, training-only feature selection and evaluation of both sensitivity and specificity. The present contribution remains independent: a characterisation of the dataset, descriptive spectral patterns and the observed limitations of the empirical-reference approach.

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